Proper ortochronous Lorentz group

$\boxed{1.}$ The 1-parameter subgroups of the Lorentz group correspond to three spatial rotations and three boosts. We can express the 1-parameter Lorentz subgroups in matrix representations along with the corresponding generators of the Lorentz algebra, where φ represents the rotation angle and u represents the rapidity associated with boosts. The generators J_x, J_y, and J_z correspond to rotations around the x-, y-, and z-axes, respectively, while the generators N_x, N_y, and N_z correspond to boosts in the x-, y-, and z-directions.

The 1-parameter subgroups can then be written as $\exp(i\varphi J)$ for rotations and $\exp(iuN)$ for boosts.

$$R_x(\varphi) = e^{i\varphi J_x} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\varphi & -\sin\varphi \\ 0 & 0 & \sin\varphi & \cos\varphi \end{pmatrix}, \quad J_x = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \end{pmatrix}.$$

$$R_y(\varphi) = e^{i\varphi J_y} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\varphi & 0 & \sin\varphi \\ 0 & 0 & 1 & 0 \\ 0 & -\sin\varphi & 0 & \cos\varphi \end{pmatrix}, \quad J_y = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}.$$

$$R_z(\varphi) = e^{i\varphi J_z} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi & 0 \\ 0 & \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad J_z = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$B_x(u) = e^{iuN_x} = \begin{pmatrix} \cosh u & \sinh u & 0 & 0 \\ \sinh u & \cosh u & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad N_x = \begin{pmatrix} 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$B_y(u) = e^{iuN_y} = \begin{pmatrix} \cosh u & 0 & \sinh u & 0 \\ 0 & 1 & 0 & 0 \\ \sinh u & 0 & \cosh u & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad N_y = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$B_z(u) = e^{iuN_z} = \begin{pmatrix} \cosh u & 0 & 0 & \sinh u \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh u & 0 & 0 & \cosh u \end{pmatrix}, \quad N_z = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}.$$

Consequently, the Lorentz group is six-dimensional.

When examining transformations in specific subspaces, such as the $t-z$ hyperplane, we observe that boost matrices transform points along hyperbolic paths, creating orbits with a hyperbolic shape. In contrast, rotation matrices in spatial subspaces (e.g., $x-y$ or $y-z$ hyperplanes) transform points along circular paths, forming compact orbits.

Both types of Lorentz transformations (rotations and boosts) preserve the spacetime interval and thus have a unit determinant (for rotations $\cos^2\varphi + \sin^2\varphi = 1$ and for boosts $\cosh^2 u - \sinh^2 u = 1$). However, their distinct behavior lies in their Hermiticity and the nature of their orbits: Rotations are unitary and generate compact orbits with closed circular paths. On the other hand, boosts are hermitian, leading to non-compact orbits with hyperbolic trajectories.

$\boxed{2.}$ In addition to the rotations and pure boosts of the Lorentz group, the Poincaré group also includes spacetime translations. We will derive the generators in a representation on functions, that is, as vector fields generating infinitesimal transformations (Killing vector fields for Minkowski spacetime).

To derive the generators of the Poincaré group, we consider an arbitrary field $f(x^\mu)$, where $x^\mu = (t, x, y, z)$ are the spacetime coordinates. The infinitesimal transformations can be expanded to first order, allowing us to identify the action of each generator.

Generators of translation

For an infinitesimal translation in the x^μ direction by a small parameter ϵ^μ, we write:

$$x'^\mu = x^\mu + \epsilon^\mu.$$

Under this translation, the function $f(x^\mu)$ transforms as:

$$f(x'^\mu) = f(x^\mu + \epsilon^\mu) \approx f(x^\mu) + \epsilon^\nu \partial_\nu f(x^\mu).$$

The term ∂_ν represents the generator of the infinitesimal translation in the x^ν direction. To include the correct convention for unitary representations, we introduce a factor of $-i$, so the generator for translation along x^μ becomes:

$$P_\mu = -i\partial_\mu.$$

Generators of rotations and boosts

For a Lorentz transformation the general relation is in the form

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu},$$

where for the Lorentz transformation matrix $\Lambda^{\mu}{}_{\nu}$ to preserve the metric, the orthogonality relations must hold:

$$\eta_{\alpha\beta} = \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}\eta_{\mu\nu}.$$

Expanding $\Lambda^{\mu}{}_{\nu}$ in terms of an identity matrix plus an infinitesimal parameter, we write:

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu}.$$

Substituting this into the metric preservation condition and keeping only terms linear in $\omega^{\mu\nu}$, we get:

$$\eta_{\alpha\beta} = (\delta^{\mu}{}_{\alpha} + \omega^{\mu}{}_{\alpha})(\delta^{\nu}{}_{\beta} + \omega^{\nu}{}_{\beta})\eta_{\mu\nu} \approx \eta_{\alpha\beta} + \omega_{\alpha\beta} + \omega_{\beta\alpha}.$$

For this equation to hold, we must have $\omega_{\alpha\beta} = -\omega_{\beta\alpha}$, showing that $\omega^{\mu\nu}$ is antisymmetric.

Now, consider the effect of this transformation on the field $f(x^{\mu})$ under this infinitesimal Lorentz transformation:

$$f(x'^{\mu}) = f(x^{\mu} + \omega^{\mu}{}_{\nu}x^{\nu}) \approx f(x^{\mu}) + \omega^{\mu}{}_{\nu}x^{\nu}\frac{\partial}{\partial x^{\mu}}f(x^{\mu}).$$

Thus, the generator of an infinitesimal Lorentz transformation in the μ-ν hyperplane is given by:

$$M_{\mu\nu} = -i\left(x_{\mu}\partial_{\nu} - x_{\nu}\partial_{m}u\right).$$

This form of the generators $M_{\mu\nu}$ ensures that only the antisymmetric part of $\omega_{\mu\nu}$ is preserved in the transformation. The generators $M_{\mu\nu}$ include both rotations (when μ, ν are spatial indices) and boosts (when one of μ or ν is the time index).

To calculate the commutator $[M_{\mu\nu}, P_{\alpha}]$, we apply these operators on a test function $f(x)$:

$$[M_{\mu\nu}, P_{\alpha}]f(x) = M_{\mu\nu}(P_{\alpha}f(x)) - P_{\alpha}(M_{\mu\nu}f(x)).$$

$$M_{\mu\nu}(P_{\alpha}f(x)) = -i\left(x_{\mu}\partial_{n}u - x_{\nu}\partial_{m}u\right)\left[-i\partial_{\alpha}f(x)\right] = -\left[x_{\mu}\partial_{\nu}\partial_{\alpha} - x_{\nu}\partial_{\mu}\partial_{\alpha}\right]f(x)$$

$$\begin{aligned} P_{\alpha}(M_{\mu\nu}f(x)) &= -i\partial_{\alpha}\left[-i\left(x_{\mu}\partial_{\nu} - x_{\nu}\partial_{\mu}\right)f(x)\right] \\ &= -\left[\eta_{\mu\alpha}\partial_{\nu} + x_{\mu}\partial_{\alpha}\partial_{\nu} - \eta_{\alpha\nu}\partial_{\mu} - x_{\nu}\partial_{\alpha}\partial_{\mu}\right]f(x). \end{aligned}$$

Subtracting the two terms, we get:

$$[M_{\mu\nu}, P_{\alpha}]f(x) = i\left(\eta_{\mu\alpha}(-i\partial_{\nu}) - \eta_{\nu\alpha}(-i\partial_{\mu})\right)f(x).$$

Thus, the commutation relation is:

$$[M_{\mu\nu}, P_\alpha] = i\left(\eta_{\mu\alpha}P_\nu - \eta_{\nu\alpha}P_\mu\right).$$

Specifically for a time translation ($\alpha = 0$) and a boost M_{0i}, we have:

$$[M_{0i}, P_0] = \eta_{00}P_i - \eta_{i0}P_0 = P_i.$$

Therefore, the commutator of time-translation and a boost in the i direction is a translation in the i-direction. This result does not disturb me :).

$\boxed{3.}$ To encode the spacetime vector x^μ as a 2×2 Hermitian matrix, we write:

$$\boldsymbol{x} = x^\mu \sigma_\mu,$$

where $\sigma_\mu = (\mathbb{1}, \vec{\sigma})$ is a 4-vector of Hermitian matrices, with σ_0 being the 2×2 identity matrix and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ being the Pauli matrices. Explicitly the components of $\boldsymbol{x}$ are:

$$\boldsymbol{x} = x^0\sigma_0 + x^1\sigma_x + x^2\sigma_y + x^3\sigma_z,$$

where

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Substituting $x^\mu = (x^0, x^1, x^2, x^3)$ into the expression for $\boldsymbol{x}$, we get:

$$\boldsymbol{x} = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}.$$

Now we calculate the determinant of $\det \boldsymbol{x}$:

$$\begin{aligned} \det \boldsymbol{x} &= (x^0 + x^3)(x^0 - x^3) - (x^1 - ix^2)(x^1 + ix^2) \\ &= (x^0)^2 - (x^3)^2 - ((x^1)^2 + (x^2)^2) = \eta_{\mu\nu}x^\mu x^\nu =: \|x\|^2 \end{aligned}$$

To ensure that the transformation

$$\boldsymbol{x}' = L\boldsymbol{x}L^\dagger$$

preserves the Minkowski norm of x^μ, we require that (using results of linear algebra)

$$\det \boldsymbol{x}' = \det(L\boldsymbol{x}L^\dagger) = \det L \det \boldsymbol{x} \det\left(L^\dagger\right) = \det L \det \boldsymbol{x} (\det L)^* = |\det L|^2 \det \boldsymbol{x} \overset{!}{=} \det(\boldsymbol{x}).$$

Thus, $|\det L| = 1$.

However, to guarantee that L belongs to the group uniquely defining a Lorentz transformation without changing orientation or time direction, i.e. *the proper ortochronous Lorentz group*, we impose $\det L = 1$, so $L \in \mathrm{SL}(2,\mathbb{C})$.

To restrict to transformations that preserve both spatial orientation and the direction of time, we focus on the proper ortochronous Lorentz group $\mathrm{SO}^+(1,3)$, which excludes time-reversing and space-reflecting transformations. The group $\mathrm{SL}(2,\mathbb{C})$ is the double cover of $\mathrm{SO}^+(1,3)$, meaning each element in $\mathrm{SO}^+(1,3)$ corresponds to two elements in $\mathrm{SL}(2,\mathbb{C})$: L and $-L$. This double covering can be easily seen as both L and $-L$ in $\mathrm{SL}(2,\mathbb{C})$ yield the same transformation when we use the relation $\boldsymbol{x}' = L\boldsymbol{x}L^\dagger$.

$\boxed{4.}$ This is a somewhat hand-waving derivation aimed at finding the matrices L corresponding to rotations and boosts in the Lorentz group. To ensure that the transformation $\boldsymbol{x}' = L\boldsymbol{x}L^\dagger$ corresponds to the Lorentz transformation:

$$Lx^\nu \sigma_\nu L^\dagger = \Lambda^\mu{}_\nu x^\nu \sigma_\mu,$$

we want this relation to hold for any spacetime vector x^ν. We begin by considering an infinitesimal Lorentz transformation

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu,$$

where $\omega^\mu{}_\nu$ is an infinitesimal antisymmetric parameter. For L, we assume an infinitesimal form:

$$L = I + M,$$

where M will contain the information about generators of Lorentz transformations in the representation on $\mathrm{SL}(2,\mathbb{C})$.

Applying this to the left side of the transformation condition, we have

$$(I+M)\sigma_\nu(I+M)^\dagger = (\delta^\mu{}_\nu + \omega^\mu{}_\nu)\sigma_\mu.$$

Expanding to first order, we get:

$$\sigma_\nu + M\sigma_\nu + \sigma_\nu M^\dagger = \sigma_\nu + \omega^\mu{}_\nu \sigma_\mu.$$

Now, subtracting the identical terms σ_ν from both sides, we find

$$M\sigma_\nu + \sigma_\nu M^\dagger = \omega^\mu{}_\nu \sigma_\mu.$$

To satisfy the transformation condition, we separate M into two cases: hermitian generators for rotations, which we label $\mathcal{J}_i$, and anti-hermitian generators for boosts, labeled $\mathcal{N}_i$. Here we use the insight from the first exercise to write:

$$M = i\mathcal{J} \quad \text{and} \quad M = i\mathcal{N}.$$

For rotations, using the hermiticity of $\mathcal{J}_i$, we assume a rotation around the i-axis with an infinitesimal angle φ. The corresponding $\omega_{\mu\nu}$ parameter for rotations around the i-axis is $\omega_{ij} = -\varphi\varepsilon_{ijk}$ with other elements zero. The commutation relation then becomes:

$$i[\mathcal{J}_i, \sigma_j] = -\varphi\varepsilon_{ijk}\sigma_k.$$

This is satisfied if we choose $\mathcal{J}_i = \frac{1}{2}\varphi\sigma_i$, giving the correct infinitesimal transformation for rotations.

For boosts, we use the anti-hermiticity of $\mathcal{N}_i$ and assume a boost in the i-direction with rapidity u. The corresponding $\omega_{\mu\nu}$ parameter for boosts in the i-direction is $\omega_{\mu\nu} = u(\delta_{\mu 0}\delta_{\nu i} + \delta_{\mu i}\delta_{\nu 0})$. The anticommutator condition then becomes:

$$i\{\mathcal{N}_i, \sigma_j\} = u\delta_{ij}\sigma_0 \quad \text{and} \quad i\{\mathcal{N}_i, \sigma_0\} = u\sigma_i.$$

This condition is satisfied if we take $\mathcal{N}_i = -\frac{1}{2}iu\sigma_i$.

Using these generators, we can now write the explicit L-matrices for rotations and boosts as exponentials of the Pauli matrices:

Rotation around the axis i by an angle φ:

$$L_i^R(\varphi) = e^{i\frac{\varphi}{2}\sigma_i}.$$

The explicit forms of the 2×2 complex matrices for each rotation are:

$$L_x^R(\varphi) = e^{i\frac{\varphi}{2}\sigma_x} = \begin{pmatrix} \cos\frac{\varphi}{2} & i\sin\frac{\varphi}{2} \\ i\sin\frac{\varphi}{2} & \cos\frac{\varphi}{2} \end{pmatrix}$$

$$L_y^R(\varphi) = e^{i\frac{\varphi}{2}\sigma_y} = \begin{pmatrix} \cos\frac{\varphi}{2} & \sin\frac{\varphi}{2} \\ -\sin\frac{\varphi}{2} & \cos\frac{\varphi}{2} \end{pmatrix}$$

$$L_z^R(\varphi) = e^{i\frac{\varphi}{2}\sigma_z} = \begin{pmatrix} e^{i\frac{\varphi}{2}} & 0 \\ 0 & e^{-i\frac{\varphi}{2}} \end{pmatrix}$$

Boosts in the direction i with rapidity u:

$$L_i^B(u) = e^{\frac{u}{2}\sigma_i}.$$

The explicit forms of the 2×2 complex matrices for each boost are:

$$L_x^B(u) = e^{\frac{u}{2}\sigma_x} = \begin{pmatrix} \cosh\frac{u}{2} & \sinh\frac{u}{2} \\ \sinh\frac{u}{2} & \cosh\frac{u}{2} \end{pmatrix}$$

$$L_y^B(u) = e^{\frac{u}{2}\sigma_y} = \begin{pmatrix} \cosh\frac{u}{2} & -i\sinh\frac{u}{2} \\ i\sinh\frac{u}{2} & \cosh\frac{u}{2} \end{pmatrix}$$

$$L_z^B(u) = e^{\frac{u}{2}\sigma_z} = \begin{pmatrix} e^{\frac{u}{2}} & 0 \\ 0 & e^{-\frac{u}{2}} \end{pmatrix}$$

$\boxed{5.}$ To determine whether a universal matrix C exists such that the complex conjugate of any L-matrix from our derived representations can be transformed by C into the original representation, we assume that such a matrix C does exist. Then, for all L in the representation, we would have:

$$L^\star = CLC^{-1}.$$

This would imply that the complex conjugate representation is equivalent to the original representation, meaning we could always transform between L and L^* via the same matrix C.

We can check if this assumption holds by examining the trace of this relation. Since the trace of a matrix product is invariant under cyclic permutations, taking the trace on both sides yields:

$$\mathrm{Tr}(L)^* = \mathrm{Tr}(L^\star) \stackrel{!}{=} \mathrm{Tr}(CLC^{-1}) = \mathrm{Tr}(L).$$

Thus, for all L in the representation, $\mathrm{Tr}(L)$ must be real.

However, let us construct a counterexample by considering a simple combination of a boost in the z-direction with rapidity u and a rotation around the z-axis by an angle φ. The corresponding L-matrix is:

$$L = \begin{pmatrix} e^{\frac{u}{2}+i\frac{\varphi}{2}} & 0 \\ 0 & e^{-\frac{u}{2}-i\frac{\varphi}{2}} \end{pmatrix}.$$

This matrix lies in $\mathrm{SL}(2,\mathbb{C})$ since $\det(L) = 1$, but its trace is:

$$\mathrm{Tr}(L) = e^{\frac{u}{2}+i\frac{\varphi}{2}} + e^{-\frac{u}{2}-i\frac{\varphi}{2}},$$

which is, in general, complex and not necessarily real. Since we have found an example where the trace is complex, the assumption that there exists a matrix C such that $\mathrm{Tr}(L^\star) = \mathrm{Tr}(L)$ for all L is false. Therefore, no such matrix C exists, meaning the complex conjugate representation is inequivalent to the original representation.

$\boxed{6.}$ In this problem, we are asked to identify a representation of the Lorentz group that is isomorphic to the vector representation we initially considered. In the case of the rotation group SO(3), we encoded vectors as traceless, Hermitian 2×2

matrices $\boldsymbol{x} = x^i\sigma_i$ (where σ_i are the Pauli matrices), and the conjugation by SU(2) matrices then provided an adjoint representation of SO(3) via SU(2). This worked because $\boldsymbol{x}$ was traceless, so it belonged to the Lie algebra of SU(2).

For the Lorentz group, however, $x^\mu\sigma_\mu$ is Hermitian but not traceless (because σ_0 is the identity matrix), meaning the vector x^μ cannot lie in the Lie algebra of $\mathrm{SL}(2,\mathbb{C})$, which consists of traceless matrices. Therefore, the vector representation of the Lorentz group $\mathrm{SO}(1,3)$ cannot be directly isomorphic to the adjoint representation of $\mathrm{SL}(2,\mathbb{C})$.

We can still analyze this problem by considering the dimensions of the representations involved. The vector representation of the Lorentz group has four real dimensions, corresponding to the four components of x^μ in Minkowski space. The complex group $\mathrm{SL}(2,\mathbb{C})$ is three complex-dimensional (or six real-dimensional) and serves as a double cover of $\mathrm{SO}^+(1,3)$, meaning each Lorentz transformation is represented by two elements L and $-L$ in $\mathrm{SL}(2,\mathbb{C})$.

To see the structure of this representation in the complexified sense, we note from exercise 4 that when considering complex rapidities or angles, we essentially work with elements of $\mathrm{SL}(2,\mathbb{C})$, where we identified that only three of the original six real generators in the Lorentz algebra are linearly independent.

When we complexify the Lorentz group, this becomes clearer. The complexified Lie algebra $\mathfrak{so}(1,3)^{\mathbb{C}}$ decomposes as:

$$\mathfrak{so}(1,3)^{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}).$$

This decomposition can be derived by defining complex combinations of the original real Lorentz generators, specifically by defining new generators for the complexified algebra:

$$L_i = J_i + iN_i, \quad R_i = J_i - iN_i,$$

where J_i are the rotation generators and N_i are the boost generators. These combinations satisfy the commutation relations of two independent $\mathfrak{sl}(2,\mathbb{C})$ algebras:

$$[L_i, L_j] = i\epsilon_{ijk}L_k, \quad [R_i, R_j] = i\epsilon_{ijk}R_k, \quad [L_i, R_j] = 0.$$

Thus, $\mathfrak{so}(1,3)^{\mathbb{C}}$ decomposes into two independent algebras, each isomorphic to $\mathfrak{sl}(2,\mathbb{C})$. These algebras correspond to two two-dimensional spinor representations of $\mathrm{SL}(2,\mathbb{C})$: the left Weyl spinor (associated with L_i) and the right Weyl spinor (associated with R_i).

Therefore, in the complexified sense, the vector representation of $\mathrm{SO}(1,3)$ is isomorphic to the tensor product of the left and right Weyl spinor representations, each corresponding to a copy of $\mathrm{SL}(2,\mathbb{C})$:

$$\mathrm{SO}(1,3)^{\mathbb{C}} \cong \mathrm{SL}(2,\mathbb{C})_L \otimes \mathrm{SL}(2,\mathbb{C})_R.$$